

**SAFETY DATA SHEET**

Creation Date 09-February-2011

Revision Date 29-March-2024

Revision Number 4

**1. Identification**

**Product Name** Magnesium acetate tetrahydrate

**Cat No. :** 36358

**CAS-No** 16674-78-5  
**Synonyms** Acetic acid, magnesium salt

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

**Details of the supplier of the safety data sheet****Company****Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887**2. Hazard(s) identification****Classification****WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)**Combustible Dusts** Category 1**Label Elements****Signal Word**

Warning

**Hazard Statements**

May form combustible dust concentrations in air

**Precautionary Statements****Prevention**

Keep container tightly closed

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

**Response**

In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion

**Storage**

Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component                        | CAS-No     | Weight % |
|----------------------------------|------------|----------|
| Acetate, magnesium, tetrahydrate | 16674-78-5 | >95      |
| Magnesium acetate                | 142-72-3   | -        |

### 4. First-aid measures

|                                        |                                                                                                                         |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Inhalation</b>                      | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable.                                                                                            |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                   |

### 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

**Specific Hazards Arising from the Chemical**

Dust can form an explosive mixture with air. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. Fine dust dispersed in air may ignite.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Magnesium oxides.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA****Health**  
1**Flammability**  
1**Instability**  
1**Physical hazards**  
N/A**6. Accidental release measures**

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions** Should not be released into the environment.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**7. Handling and storage**

**Handling** Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

**Storage.** Store under an inert atmosphere. Keep container tightly closed in a dry and well-ventilated place. Protect from moisture. Incompatible Materials. Strong oxidizing agents. Strong acids. Strong bases.

**8. Exposure controls / personal protection**

**Exposure Guidelines** This product does not contain any known or suspected reproductive hazards

**Engineering Measures** None under normal use conditions.

**Personal protective equipment**

**Eye Protection** Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection** Protective gloves

**Glove material**

Nitrile rubber  
Neoprene  
Natural rubber  
PVC

**Breakthrough time**

See manufacturers  
recommendations

**Glove thickness**

-

**Glove comments**

Splash protection only

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particle filter

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do

not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Solid                    |
| Appearance                             | White                    |
| Odor                                   | vinegar-like             |
| Odor Threshold                         | No information available |
| pH                                     | 8-9 5% aq.sol            |
| Melting Point/Range                    | 80 °C / 176 °F           |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | Not applicable           |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | Not applicable           |
| Specific Gravity                       | 1.454 (H2O=1)            |
| Solubility                             | Soluble in water         |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition Temperature              | No information available |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | C4 H6 O4 Mg . 4 H2 O     |
| Molecular Weight                       | 214.46                   |

## 10. Stability and reactivity

|                                  |                                                                                                                             |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                                  |
| Stability                        | Hygroscopic.                                                                                                                |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Heat, flames and sparks. Exposure to moisture. Exposure to moist air or water. |
| Incompatible Materials           | Strong oxidizing agents, Strong acids, Strong bases                                                                         |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Magnesium oxides                                                   |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                    |
| Hazardous Reactions              | None under normal processing.                                                                                               |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Mist LC50   | Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.     |

#### Component Information

| Component         | LD50 Oral                 | LD50 Dermal                   | LC50 Inhalation |
|-------------------|---------------------------|-------------------------------|-----------------|
| Magnesium acetate | LD50 = 8610 mg/kg ( Rat ) | LD50 > 20000 mg/kg ( Rabbit ) | Not listed      |

Toxicologically Synergistic Products No information available

Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                        | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------------|------------|------------|------------|------------|------------|------------|
| Acetate, magnesium, tetrahydrate | 16674-78-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Magnesium acetate                | 142-72-3   | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

DOT Not regulated

TDG Not regulated

IATA Not regulated

IMDG/IMO Not regulated

## 15. Regulatory information

### International Inventories

| Component | CAS-No | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|-----------|--------|-----|------|------|-----------------------------------------------|--------|--------|-----|
|-----------|--------|-----|------|------|-----------------------------------------------|--------|--------|-----|

|                                  |            |   |   |   |        |           |   |   |
|----------------------------------|------------|---|---|---|--------|-----------|---|---|
| Acetate, magnesium, tetrahydrate | 16674-78-5 | - | - | - | -      | -         | - | - |
| Magnesium acetate                | 142-72-3   | X | - | X | ACTIVE | 205-554-9 | - | - |

| Component                        | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Acetate, magnesium, tetrahydrate | 16674-78-5 | X     | -        | -    | -    | X    | X    | X     | X     |
| Magnesium acetate                | 142-72-3   | X     | KE-22674 | X    | X    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

INECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

Authorisation/Restrictions according to EU REACH

Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                        | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Acetate, magnesium, tetrahydrate | 16674-78-5 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |
| Magnesium acetate                | 142-72-3   | Listed         | Not applicable               | Not applicable            | Not applicable                             |

| Component                        | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Acetate, magnesium, tetrahydrate | 16674-78-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Magnesium acetate                | 142-72-3   | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information****Prepared By**

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the

date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**